12. The Rings

A ring breaks, grinds and returns: the source of the material, its dynamics, and the marks it leaves on moons and on the parent planet

Zohar Duran · ORCID 0009-0005-3194-9387 · 10 November 2026

Abstract

The research series has established that a planetary ring is not a site of accretion that builds moons but a mechanism of grinding and disintegration [Paper 11]. This paper extends that principle into a unified physical and dynamical model of ring systems, defining the source of the material, the conditions under which a ring is sustained, the mechanisms by which its mass moves, and the morphological signatures left on the moons and on the parent planet.

1. The source of ring mass: all the material that makes up a planetary ring originates in the parent body, through two mechanical pathways:

- **Direct feeding:** material ejected from the parent's surface (lava, water and vapour) that did not cross the corotation line; this mass remains trapped in orbit, is ground and dispersed without condensing into a moon.
- **Secondary feeding:** material ejected from moons that formed from the parent and continue to feed the system (captured moons are excluded from this rule).

Saturn's rings – made of 95%–99% water ice, with a total mass of 1.54×10^{19} kg in a thin equatorial disc – originate in the planet's interior and are not a product of captured interplanetary dust.

2. The ring existence equation (dynamic equilibrium): a ring system exists as long as it receives a continuous supply of mass. At Saturn a mass-removal rate ("ring rain") of $432 - 45,000$ kg/s into the planet has been measured; without a supply of material, the ring system would disintegrate and vanish within 11 million to 1.1 billion years. The ring is therefore a steady mass flow in equilibrium, fed from within, at a cumulative cost of up to about one part in a hundred thousand (1.1×10^{-5}) of Saturn's mass. On Mars, the shutdown of the hydrodynamic pump and the drying of the moons led to the ring's disappearance. A ring with neither direct feeding nor feeding by a moon (such as the

narrow moonless rings of Uranus and the Galle and Lassell rings of Neptune) is doomed to fade away.

3. The two conditions for the formation of a massive ring:

- **A dynamical-geometric condition:** the corotation line must lie below the Roche limit ($a_c < a_R$). Comparing the two equations cancels the parent's density and yields a critical rotation period that is the same for all celestial bodies: $P^* \approx 13.4 \text{ hours}$ (requiring $P < 13.4 \text{ hours}$).
- **A thermodynamic-structural condition:** active leakage of material from the surface or the envelope of the parent planet.

Only present-day Saturn and ancient Mars meet both criteria at once. Jupiter meets the speed condition ($P = 9.9 \text{ hours}$), but its thick gas envelope blocks the leakage of water; Uranus and Neptune are too slow and froze. The rings of the gas and ice giants lie below or around the corotation line, whereas the rings of the small bodies lie above it, near the radius of the 1:3 orbital resonance.

4. Loss of angular momentum through the escape of water: on Uranus and Neptune the corotation line migrated radially outward. The present moons account for only 1% (Uranus) and 0.002% (Neptune) of the angular momentum the planets lost. The water that escaped through the corotation line weighed 270 times the mass of all the moons of Uranus and was ejected into outer space. This massive loss of water stems from the thin atmospheric cap and the low escape speed of the ice giants, not from temperature conditions.

5. Morphological signatures on the moons: an icy ring causes mass to be accreted (the formation of equatorial ridges on Pan, Atlas and Daphnis), whereas a silicate ring causes mechanical grinding (Deimos stayed for a long time in the heart of the Martian ring and was ground, while Phobos crossed its lower edge and was ground only partly). No ring has been found around any moon in the Solar System.

The quantitative requirements, the predictions and the refutation criteria are collected in Part L.

`What this paper proposes`

• **Central claim:** a planetary ring is an intermediate stage in a cyclic mass flow that leaves the parent planet and returns to it. All of the ring's material originates in the parent (directly from the ground or through original moons), and the ring stays stable as long as the supply of material continues. The ring does not produce moons, but it reshapes the morphology of the moons that cross it. A massive ring forms only when the parent's rotation period is shorter than 13.4 hours and the inner crust leaks.

• Main equations and parameters:

• **Position of the corotation line:** $a_c = \left(\frac{GM}{\Omega^2}\right)^{1/3}$

• **Roche limit for ice particles:** $a_R \approx 2.46R\left(\frac{\rho}{\rho_{\text{ice}}}\right)^{1/3}$

• **Critical rotation period for a massive ring:** $P^* \approx 13.4 \text{ hours}$

• **Ring-rain rate at Saturn:** $432 - 45,000 \text{ kg/s}$

• **Mass of Saturn's rings:** $1.54 \times 10^{19} \text{ kg}$

Working environment

Physical quantity	Measured / computed value	Source
Mass of Saturn's rings	1.54×10^{19} kg (about 40% of the mass of Mimas)	Iess et al. 2019
Composition of Saturn's A and B rings	95%–99% water ice	Cuzzi et al. 2010
Vertical thickness of the rings	dense rings: about $10 - 30$ m; E ring: thousands of kilometres	Cuzzi et al. 2010
Neutral ring rain (to the equator)	$4,800 - 45,000$ kg/s	Waite et al. 2018
Ionised ring rain (along the field lines)	$432 - 2,870$ kg/s	O'Donoghue et al. 2019
Corotation line (in planet radii)	Jupiter: $2.29R$; Saturn: $1.92R$; Uranus: $3.26R$; Neptune: $3.39R$	Calculation [Part A.3]
Median depth-to-width ratio of depressions	Deimos: 1.9%; Phobos: 5.5%	Ernst et al. 2023; Paper 10

PART A Introduction – two lines around every planet

A.1 The research question

All four gas and ice giants in the Solar System have ring systems, but their physical properties differ completely: Saturn's rings are massive and of high albedo, Jupiter's rings are tenuous and made of dust, and the rings of Uranus and Neptune are narrow and dark. In addition, three small bodies at the edge of the Solar System (Haumea, Chariklo, Quaoar) have dense rings, one of which lies far beyond the Roche limit. The inner planets have no rings at present, but Mars had an ancient ring system [Paper 10; Ćuk et al. 2020]. No ring has been found around any moon in the Solar System.

Paper 11 established the corotation line as the axis around which moons form, and defined a ring as a site of mechanical grinding and disintegration. This part examines how the geometric and dynamical interplay between the corotation line and the Roche limit determines a ring's position, its chemical composition, its lifetime and its structural effect on the bodies that cross it.

A.2 The corotation line and the Roche limit

- **The corotation line (synchronous radius):** the radial distance from the centre of the parent planet at which an orbiting body completes one orbit in a time equal to the planet's rotation period:

$$a_c = \left(\frac{GM}{\Omega^2}\right)^{1/3}$$

- Below the line ($a < a_c$): the orbital angular speed is higher than the parent's rotation rate. The tidal bulge raised in the planet lags behind the orbiting body, exerts a braking force on it and forces it to migrate radially inward.
- Above the line ($a > a_c$): the orbital angular speed is lower than the parent's rotation rate. The tidal bulge leads the body, accelerates it and pushes it radially outward [Paper 11, A.6].
- The Roche limit (a_R): the radial distance below which the parent's differential tidal forces overcome the self-gravity of a soft body and prevent hydrostatic condensation. For water-ice particles the limit is given by the equation:

$$a_R \approx 2.46\, R \left(\frac{\rho}{\rho_{\text{ice}}}\right)^{1/3}$$

where R is the planet's radius, ρ its mean density, and ρ_{ice} the density of ice.

A.3 Where the rings sit

Normalising the positions of the rings and of the Roche limit to the corotation radius (a_c) allows a direct comparison between bodies of different masses and rotation rates on one uniform dynamical scale:

Parent body	Corotation line a_c (km)	Ring range (km)	Ring positions in units of a_c	Roche limit for ice in units of a_c	Position relative to the line
Jupiter	160,000	92,000–226,000 (halo, main, gossamer)	$0.57 - 1.41$	1.25	Around the line
Saturn	111,600	66,900–136,800 (D to A rings)	$0.60 - 1.23$	1.21	Around the line; outer edge at the Roche limit
Uranus	82,700	41,800–51,100	$0.51 - 0.62$	0.85	Below the line
Neptune	83,500	41,900–62,900	$0.50 - 0.75$	0.89	Below the line
Haumea	1,104	2,287	2.07	2.27	Above the line
Chariklo	186	391; 405	2.10; 2.18	1.71	Above the line
Quaoar	2,020	2,520; 4,057	1.25; 2.01	0.83	Above the line; both outside the Roche limit

The data reveal a clear division into two separate dynamical families (Figures A.1 and A.2):

1. The family of the gas and ice giants: the rings lie below the corotation line or cross it ($0.50 - 1.41\ a_c$), mostly inside the Roche limit; the part below the line migrates inward and falls onto the parent planet. At Saturn, the outer edge of the A ring ($136,800\ \mathrm{km}$) coincides almost exactly with the Roche limit for ice ($135,300\ \mathrm{km}$).

2. The family of the small bodies (dwarf planets and centaurs): the rings lie far above the corotation line ($1.25 - 2.18\ a_c$), in the region where the tidal vector pushes mass outward. Most of them cluster near $2.08\ a_c$ – the exact radius of the 1:3 spin-orbit resonance, where a ring particle completes one orbit while the parent body completes three rotations about its axis [Sicardy et al. 2019].

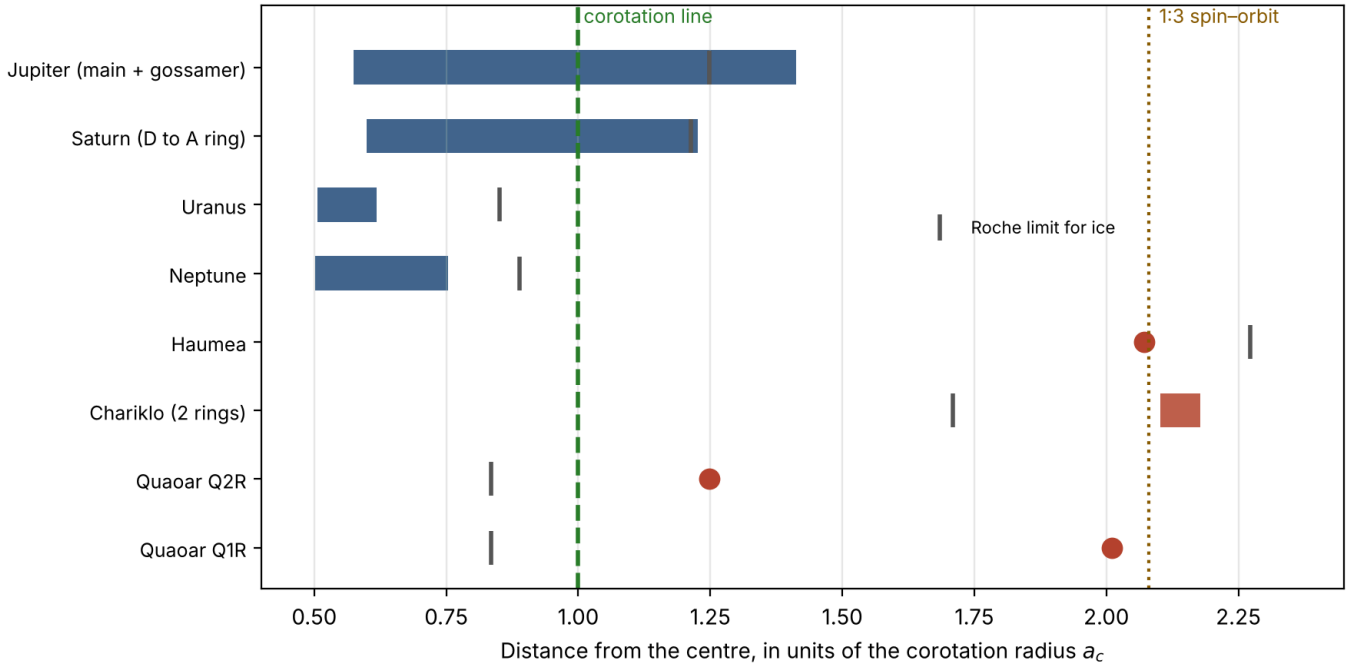
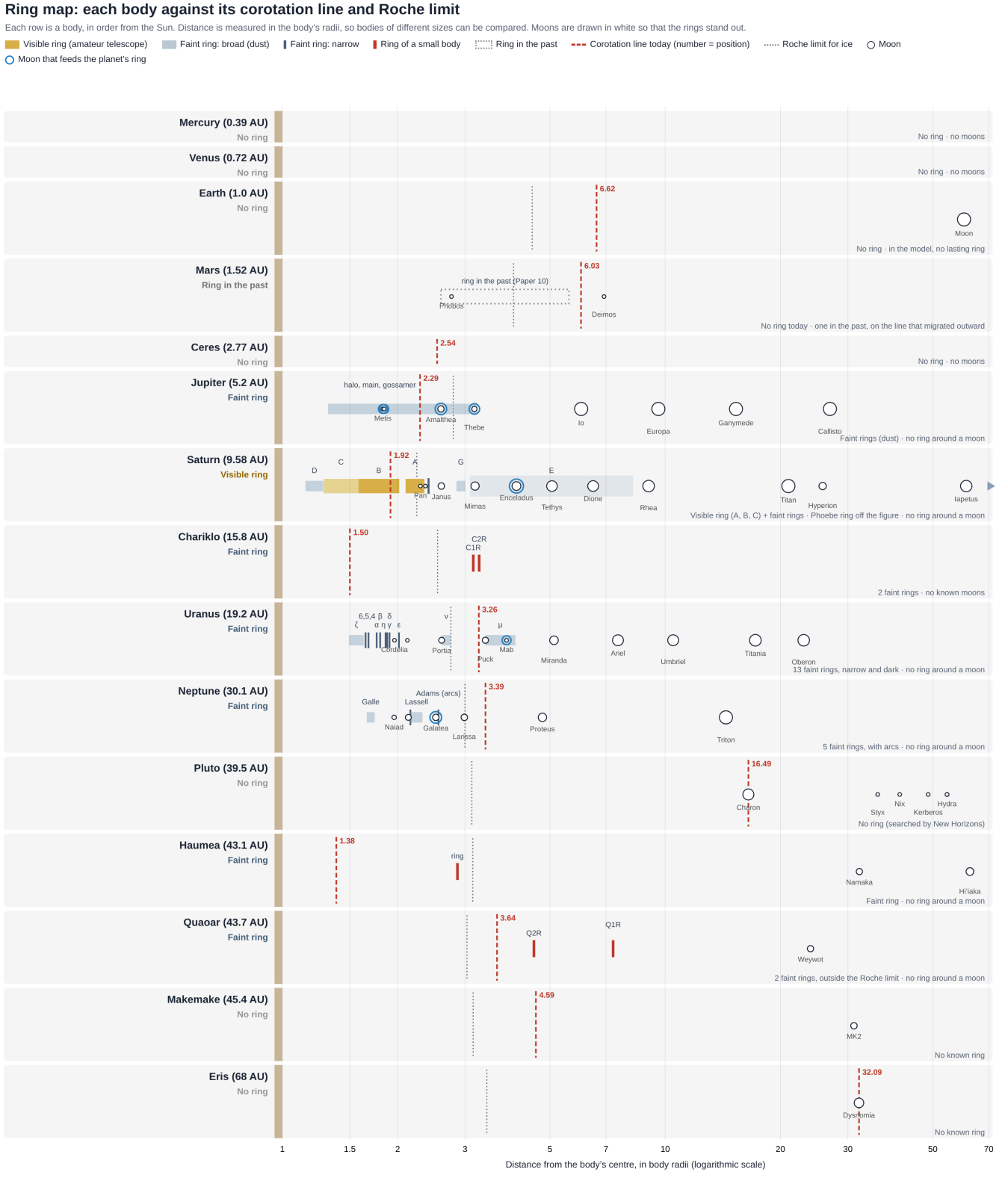


Figure A.1. Position of the rings in units of the corotation radius. Blue: giant planets; red: small bodies. Green dashed line: the corotation line. Brown dotted line: the 1:3 resonance. Short grey marks: the Roche limit for ice. Saturn's E ring, fed by Enceladus, is not shown. *How to read:* a ring to the left of the green line lies where material falls into the planet; a ring to the right lies where material is pushed outward.



Ring around a moon: none has been found around any moon. The suspected ring of Rhea was ruled out in 2010. Moons only feed the planet's rings (blue circles): Enceladus the E ring, Metis/Adrastea/Amalthea/Thebe the rings of Jupiter, Mab the μ ring of Uranus; Galatea holds Neptune's arcs; Phoebe feeds a huge ring off the figure (arrow in Saturn's row).

What it shows: at the giant planets the rings lie before the corotation line or around it, inside the Roche limit. At the small bodies (Chariklo, Haumea, Quaoar) the rings lie beyond the line. Mars: a ring in the past around the line, which has since migrated out to 6.03.

Figure A.2. Ring map: each body, in order from the Sun, against its corotation line (red dashed) and its Roche limit for ice (grey dotted). Distance in body radii, on a logarithmic scale. Gold: a ring visible in an amateur telescope; blue-grey: a faint ring; red: a ring of a small body; dotted frame: a ring in the past. A moon that feeds a ring is circled in blue. No ring has been found around any moon.

PART B The source of the material

B.1 Leakage from within

Following the model developed in Paper 11, differentiation drives water and light volatile components upward from the depths of the planet (the pump principle), while the atmospheric gas envelope confines them (the cap principle). This confinement is not hermetic: part of the mass leaks to the region of the corotation line, is trapped inside the Roche limit and feeds the ring system. This mechanism ensures continuous renewal of the ring as long as differentiation continues and the outlet remains active.

In the ancient Martian system, the ring was fed through the volcanic outlet on the protrusion. As long as the crust had a low viscosity and allowed volatiles to escape, the ring was fed with a large mass. As the rotation slowed, sorting advanced and the cold crust thickened, the pump's outlet was blocked, the feeding stopped and the ring faded away [Paper 10].

B.2 Ruling out an interplanetary dust source

- **Chemical purity:** Saturn's A and B rings consist of 95%–99% pure water ice [Cuzzi et al. 2010]. Interplanetary dust is a dark mixture of silicates, carbon and metals; had the rings been built of space dust, their albedo would have been low from the moment they formed.
- **An external contamination mechanism:** Cassini data measured directly the rate at which micrometeoroids fall onto the rings [Kempf et al. 2023]. Space dust contaminates an existing ring and is not its building material. The albedo gap between the dark C ring and the bright A and B rings arises from the same contamination rate: a ring of low mass darkens faster than a dense ring [Cuzzi & Estrada 1998].
- **Mass distribution and geometry:** a mass of $1.54 \times 10^{19}\ \mathrm{kg}$ is concentrated in a thin equatorial disc whose vertical thickness is tens of metres. Dust entering the system from all directions (isotropically) carries no common angular-momentum vector and cannot settle into a single flat equatorial disc; an internal source rotating in synchrony with the parent planet is required.
- **Correlation with the thickness of the atmospheric cap:** at Saturn, the thick gas envelope confines the water and produces an almost entirely pure ice ring. At Uranus and Neptune the cap is thin and the water escaped through the corotation line [Paper 11, A.9], leaving narrow, dark rings. At Jupiter, where the cap is especially thick, the rings consist of tenuous dust fed by the inner moons alone [Owen et al. 1979]. On Mars, the ring consisted of condensed vapour and silicate material from the crust [Paper 10].

B.3 A moon that feeds a ring

Saturn's E ring is fed directly by jets of water vapour and ice ejected from Enceladus. At Jupiter, the inner moons Metis, Adrastea, Amalthea and Thebe release the dust that feeds the ring system. The material flows in one direction only: a moon feeds a ring, and a ring does not produce a moon.

B.4 Two feeding pathways and the condition for sustaining a ring

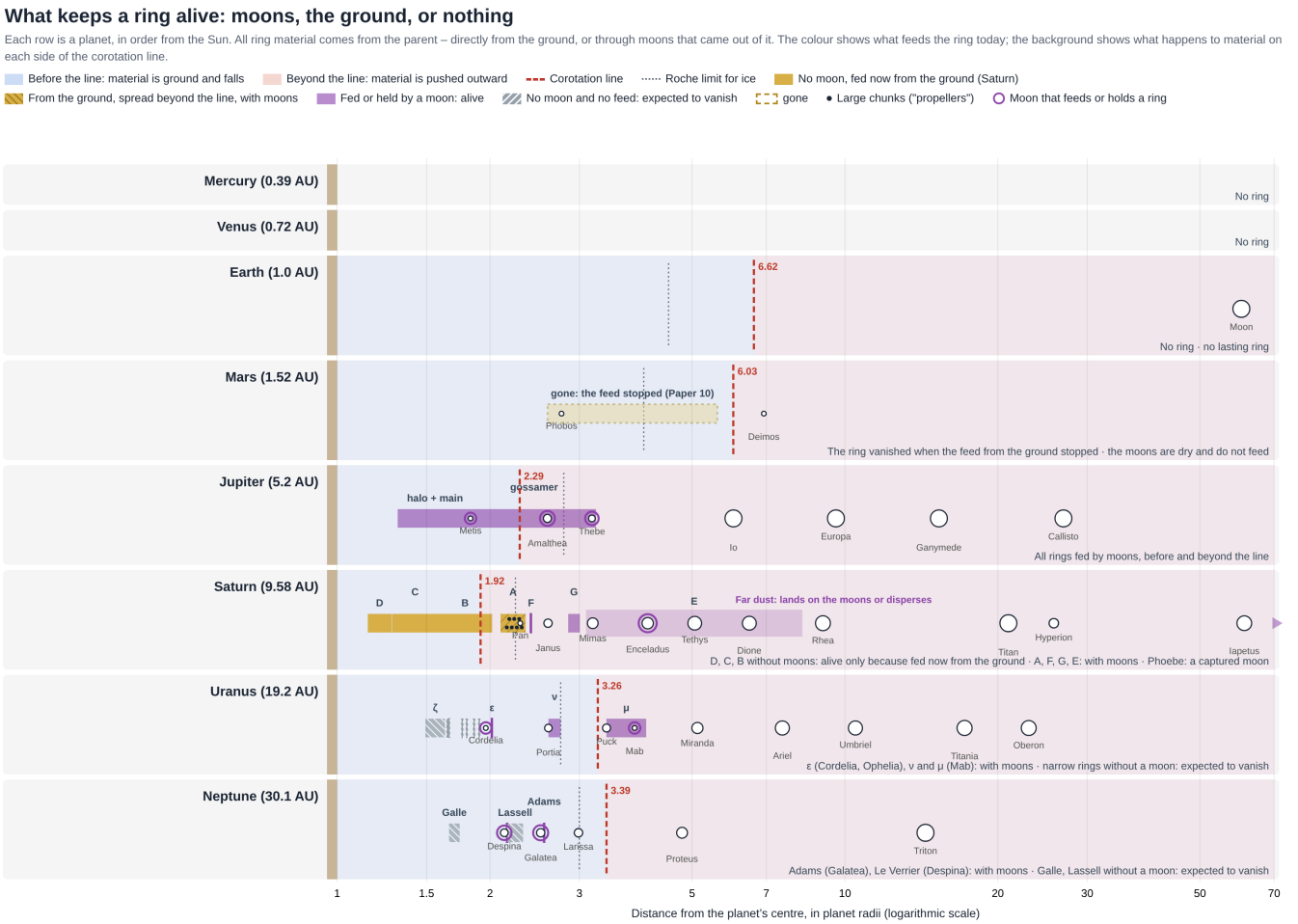
All the material that makes up ring systems leaves the parent body by two mechanical pathways:

1. Direct feeding: material ejected from the parent's surface (lava, water, vapour and dust) that remains below the corotation line. This material is trapped in orbit, ground and dispersed without condensing into a moon. Saturn's D, C and B rings (which have no moons) are fed by this pathway, and so was the ancient Martian ring [Paper 10].

2. Feeding through moons: original moons born from the parent that continue to eject mass. Enceladus feeds the E ring with jets of water; Metis, Adrastea, Amalthea and Thebe feed Jupiter's rings with dust splashed off by micrometeoroid impacts [Burns et al. 1999]; the moon Mab feeds the μ ring of Uranus.

A ring persists as long as it is fed by one of the two pathways. A moon serves as a source only if it is geologically active or sheds dust; dry, static moons (such as Phobos and Deimos) do not feed rings. When feeding from the ground stops and the moons are dry, the ring fades away – the mechanism that explains the disappearance of the Martian ring. A ring with no feeding moon and no leakage from the ground is in a post-birth stage and is expected to disappear (F.3).

- **The dynamics of distant dust dispersal:** the magnetised plasma rotating with the planet pushes dust beyond the corotation line radially outward; at large distances, solar radiation drag (the Poynting–Robertson effect) slows the particles and carries them back inward. As a result, dust from the Phoebe ring lands on the leading hemisphere of Iapetus and creates its dark side [Verbiscer et al. 2009], and dust from the E ring coats the surfaces of Mimas, Tethys, Dione and Rhea [Verbiscer et al. 2007].
- **A caveat for captured moons:** a moon captured from outside (by the scoring protocol of L.3 in Paper 11) is not part of the parent's material cycle; hence the Phoebe ring (associated with a captured moon) is excluded from the model.



The rule: all ring material comes from the parent, in two ways: directly – material ejected from the ground (lava, water, vapour) that did not cross the line; it orbits in the ring, is ground and disperses, and does not become a moon – or through moons that came out of the parent and go on releasing material. A ring lives as long as one of the two ways feeds it. When the feed from the ground stops and the moons are dry, the ring vanishes (Mars). A ring with no moon and no feed from the ground is expected to vanish. Far material disperses into space or lands on the moons.
Outside the rule: a captured moon (Phoebe) did not come from the parent; it is defined in Paper 11.

Figure B.1. What keeps a ring alive. Each row is a planet, and the colour shows what feeds the ring today:
– Gold: fed from the ground, with no moon (Saturn).
– Gold hatched: from the ground, spread beyond the line, with moons and large chunks (the A ring).
– Purple: fed or held by a moon.
– Grey hatched: no moon and no feed, expected to vanish.
– Dotted frame: gone.
Background: blue before the corotation line, where material is ground and falls; pink beyond it, where material is pushed outward.

B.5 Two conditions for the formation of a massive ring

The existence of a massive ring fed directly from the parent's ground depends on two physical criteria being met simultaneously (like a net and balls – the net must be placed in the right position, and mass must be thrown):

- Condition 1 – the corotation line lies inside the Roche limit ($a_c < a_R$): material leaking from the surface reaches the region of the corotation line (B.1). If the line lies below the Roche limit, the differential tidal forces prevent hydrostatic condensation and the mass stays as a

ring. If the line lies beyond the Roche limit, the material is not trapped as a ring. Dividing the equation of the corotation line by that of the Roche limit cancels the planet's density (ρ) completely:

$$\frac{x_c}{x_R} = \left(\frac{P}{P^*}\right)^{2/3}, \quad P^* = 2\pi\sqrt{\frac{3 \cdot 2.46^3}{4\pi G \rho_{\text{ice}}}} \approx 13.4 \text{ hours}$$

where P is the parent's rotation period (day) and $\rho_{\text{ice}} = 900 \text{ kg/m}^3$. A universal rule follows: the corotation line lies below the Roche limit if and only if the parent's day is shorter than 13.4 hours, regardless of the body's mass, radius or surface gravity (Jupiter has the strongest gravity, while Saturn's gravity is similar to Earth's).

- **Condition 2 – active leakage from the ground:** the atmospheric gas envelope does not seal the planet's interior completely, and the outer layer has not frozen through or dried thermally.

The matrix of conditions for a massive ring in the Solar System

Planet and epoch	Day P (hours)	Position of the line (R)	Roche limit for ice (R)	Condition 1 ($P < 13.4$ h)	Condition 2 (leakage from the ground)	Dynamical outcome
Saturn today	10.6	1.92	2.24	Yes	Yes: the cap leaks	Massive ring from the ground
Young Mars	7–9	1.74–2.05	2.68	Yes	Yes: a volcano on a protrusion	There was an ancient ring
Jupiter today	9.9	2.29	2.80	Yes	Almost none (thickest cap)	A dust ring from moons only
Earth at birth	5.8	2.29	4.02	Yes	No: dried quickly	No ring; the Moon was built
Uranus today	17.2	3.26	2.76	No	No: the water escaped, froze	Narrow remnants and moon rings
Neptune today	16.1	3.39	3.00	No	No: froze	Narrow remnants and moon rings
Mars today	24.6	6.03	4.02	No	No: the ground closed	The ring vanished

Earth today	23.9	6.62	4.49	No	No	No ring

• **Analysis of the matrix:** a massive ring fed from the ground develops only when both conditions are met simultaneously – Saturn today and young Mars. Jupiter's rotation meets Condition 1, but the thick atmospheric cap blocks the leakage of water. Earth at the moment of birth met Condition 1, but its thermal drying blocked the feeding of a ring. Uranus and Neptune rotate with a day of 16–17 hours (above the 13.4-hour threshold), their corotation line lies beyond the Roche limit, and the frozen crust blocks ejection. Mars shows a two-stage decline: the slowing of its rotation pushed the line beyond the Roche limit, the closing of the crust stopped the feeding, and the ring vanished. As with the arc of moon births in Paper 11, ring systems peak at Saturn, owing to the balance between avoiding drying (near the Sun) and avoiding freezing (far from the Sun).

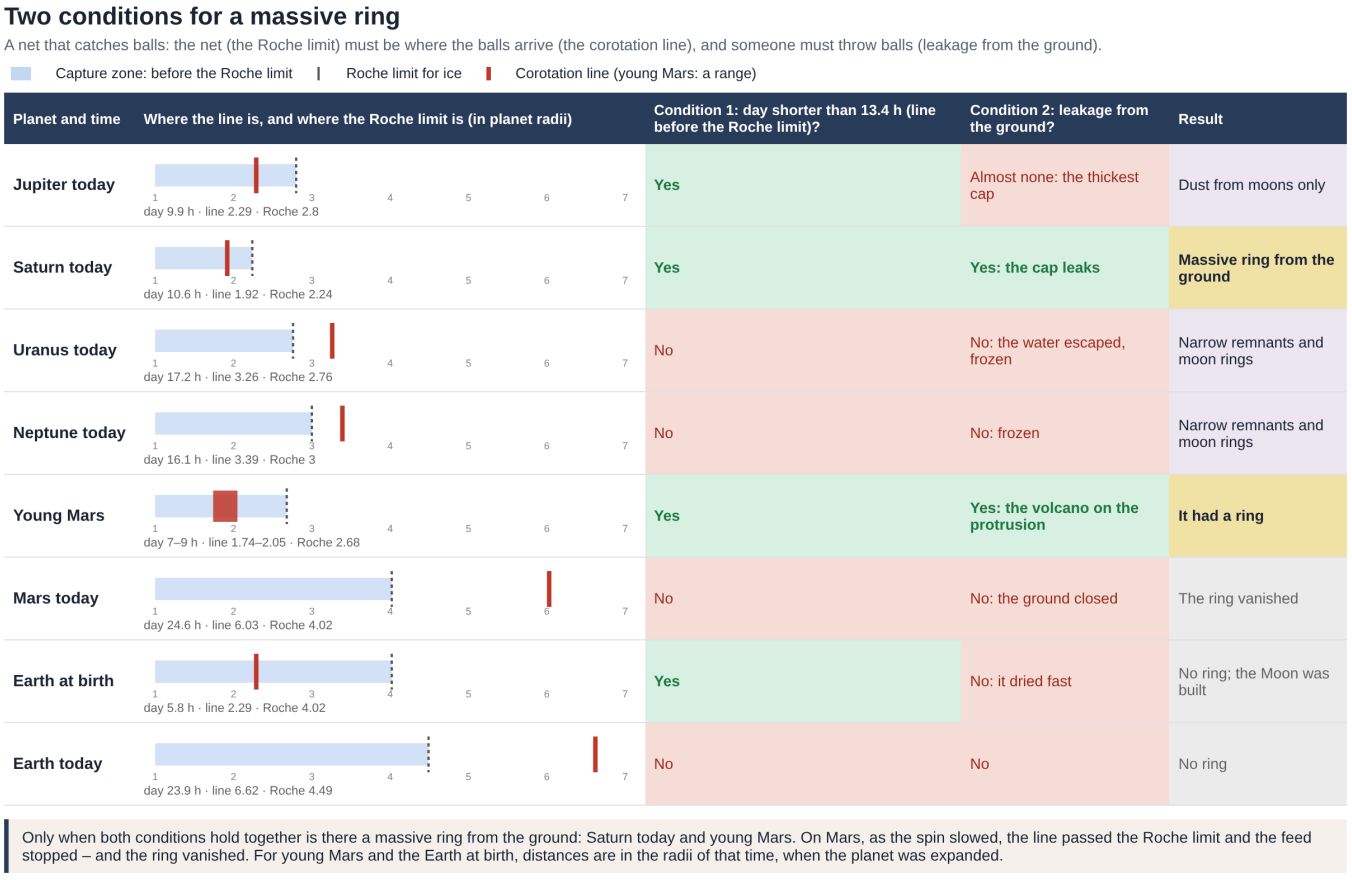


Figure B.2. Two conditions for a massive ring. Condition 1 is equivalent to a day shorter than 13.4 hours. Each row shows the capture zone (blue) up to the Roche limit (dotted line), and the position of the corotation line (red). Green: the condition holds; pink: it does not. *How to read:* if the red line lies inside the blue zone, condition 1 holds.

Figure B.3 shows the first condition across the Solar System. The corotation line today, relative to the Roche limit, forms a U: on Jupiter and Saturn the line lies inside the Roche limit (0.82 and 0.86), where the richest ring systems are. On the sides the line moved out of the ring zone – a little on Uranus and Neptune (1.18 and 1.13), where narrow rings remain, and a lot on Mars (1.50), whose ring vanished, and on Earth (1.47); on Mercury and Venus to 22 and 57 times. We propose that the line moved outward on the sides because mass left through the line – gas and water – and carried angular momentum away: on the near side because of the heat of the young Sun, on the far side because of the spin. In the middle the thick gas cap stayed, and the spin stayed fast. The calculation, planet by planet, with every parameter, is in Paper 11 (section L.1, Table L.1).

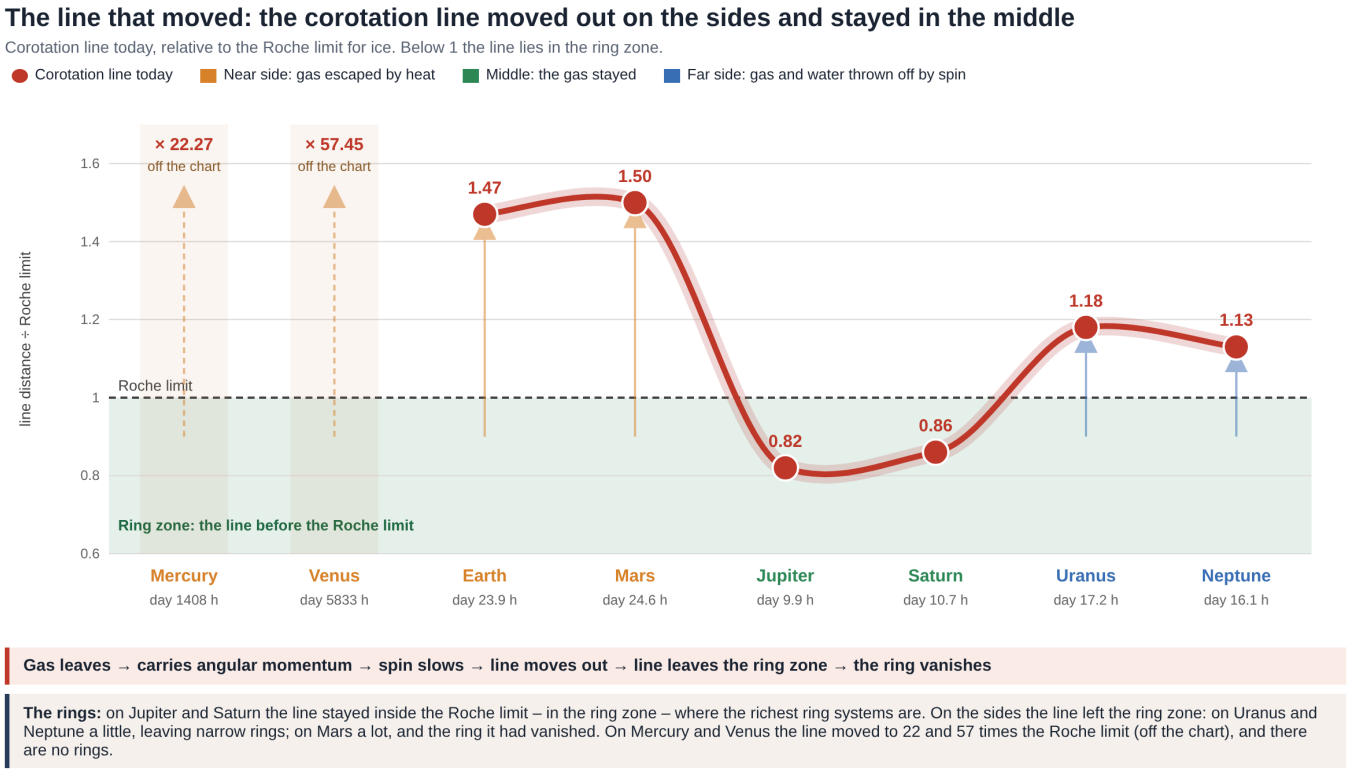


Figure B.3. The line that moved: the first condition for a ring across the Solar System. The corotation line today relative to the Roche limit for ice; below 1 the line lies in the ring zone. On Jupiter and Saturn the line stayed inside the ring zone; on the sides it left it, and on Mercury and Venus it moved to 22 and 57 times (off the chart). At the bottom: the chain in which the mass that left moves the line out of the ring zone.

PART C A ring breaks and grinds, and does not build

C.1 Dynamics inside the Roche limit

Inside the Roche limit, the parent's differential tidal stresses overcome the self-gravity of the ring particles and prevent hydrostatic condensation. Any massive body that crosses the Roche

limit inward is torn apart mechanically and ground by mutual collisions into gravel and dust. Saturn's A ring ends exactly at the Roche limit for ice, and beyond that line no moon has condensed from ring material.

C.2 Dynamics outside the Roche limit

The outer ring of the dwarf planet Quaoar lies at a distance of $4,057\ \mathrm{km}$, 2.4 times the Roche limit for ice. According to classical models, material at such a distance should condense into a moon within a few decades, yet a dense, stable ring is observed [Morgado et al. 2023]. This finding agrees fully with the unified mechanism: a ring does not produce moons, even where the parent's tidal forces do not prevent condensation. Likewise, the rings of the centaur Chariklo also lie slightly beyond the Roche limit.

C.3 The morphology of the ring moons

The tiny moons within and at the edges of Saturn's ring system – Pan, Daphnis, Atlas, Prometheus, Pandora, Janus and Epimetheus – have very low densities of $340 - 640\ \mathrm{kg/m^3}$ [Porco et al. 2007]. In line with Paper 11, these bodies are not products of ring condensation but remnants of ancient moons that were ground and polished inside the ring, onto which a layer of ring material accreted. Accretion of mass onto an existing core is not the creation of a new moon. The theoretical model of Crida and Charnoz, in which moons condense as the ring disc spreads beyond the Roche limit, rests on numerical simulation alone and is not backed by an observational finding.

C.4 The ring as a site of destruction

The model of Wisdom and colleagues, which proposes that Saturn's rings formed when an ancient moon approached below the Roche limit and disintegrated, agrees with the mechanical principle of the unified mechanism: a planetary ring functions as a site where moons are broken up and destroyed, not as a construction site.

PART D The dynamics – what returns and what leaves

D.1 Most of the material returns inward

Inside a ring disc, particles on inner orbits move with a higher angular speed than particles on outer orbits (differential Keplerian motion). Inelastic collisions between the particles convert orbital energy into thermal energy (heat), while the total angular momentum of the disc is

conserved. Satisfying both conservation laws at once forces viscous spreading: most of the ring's mass loses energy and flows inward to the parent planet, while a small part of the mass is pushed radially outward and carries the excess angular momentum with it [Lynden-Bell & Pringle 1974]. The ring functions as a dynamical funnel that returns most of the material to the parent.

D.2 The line divides

Mechanisms that act in synchrony with the planet's rotation brake material located below the corotation line and drop it inward, and push outward material located above the line. Gravitational tidal torques act on moons and massive blocks; on electrically charged dust particles acts the planetary magnetic field, which rotates together with the parent. A dense, massive ring behaves mainly as a continuous body that spreads mechanically (D.1).

- **Empirical evidence of the line acting on dust:** the "spokes" phenomenon in Saturn's ring system appears in the outer part of the B ring, in a radial range between $102,000\ \mathrm{km}$ and $112,000\ \mathrm{km}$ – a boundary that coincides exactly with the position of the corotation line ($111,600\ \mathrm{km}$) [Callos et al. 2025].
- **Transfer of angular momentum to moons at the ring edges:** the ring transfers angular momentum to the moons at its edges and pushes them radially outward [Goldreich & Tremaine 1980]. In the ancient Martian system this mechanism pushed Deimos outward, while the gaseous inner edge of the ring held Phobos on the corotation line and carried it outward together with the line, until the ring dried at a radius of about $19,000\ \mathrm{km}$ [Paper 10].

D.3 Saturn: a steady flow

Measurements by the Cassini spacecraft recorded material falling from Saturn's rings into the atmosphere through two parallel channels: neutral material flowing to the equatorial plane at a rate of $4,800 - 45,000\ \mathrm{kg/s}$ [Waite et al. 2018], and ionised material flowing along the magnetic field lines at a rate of $432 - 2,870\ \mathrm{kg/s}$ [O'Donoghue et al. 2019].

Mass fall rate ($\mathrm{kg/s}$)	Time for the ring to empty without feeding	Total feeding required over 4.5 billion years	Fraction of Saturn's total mass
432	1.1 billion years	4 ring masses	1×10^{-7}
2,870	170 million years	26 ring masses	7×10^{-7}
4,800	100 million years	44 ring masses	1.2×10^{-6}

45,000	11 million years	415 ring masses	1.1×10^{-5}

Without a supply of material, the ring system would disintegrate in a time much shorter than the age of the Solar System; this finding has led the literature to conclude that the rings are chronologically young [Iess et al. 2019; Kempf et al. 2023]. According to the unified mechanism, the ring is not a static remnant but a steady flow in dynamic equilibrium: it is renewed continuously by leakage from below and emptied from above as ring rain. The ring's "young" appearance results from the constant turnover of material, while the geodynamic process itself is ancient.

- **A mechanical requirement in the steady state:** the rate at which mass leaks from the parent body into the ring equals exactly the rate at which ring rain removes mass. The cumulative mass cost over Saturn's whole lifetime reaches at most about one part in a hundred thousand (1.1×10^{-5}) of the planet's mass, and therefore stays within the limits of mass conservation.

D.4 Where did the angular momentum go: moons or water

A radial recession of the corotation line indicates that the planet lost angular momentum. The following calculation compares the total angular momentum the parent lost since the birth stage with the orbital angular momentum stored in the regular moon system today (with moment-of-inertia factors $k \approx 0.22$ for Uranus and $k \approx 0.23$ for Neptune; the amount of water is computed for ejection from the ancient corotation line at $2R$ [Paper 11]):

Planet	Change of the day	Angular momentum lost ($\mathrm{kg \cdot m^2/s}$)	Orbital angular momentum of the moons ($\mathrm{kg \cdot m^2/s}$)	Moons' share of the loss	Mass of water required to leave through the line
Uranus	$8.3 \to 17.2$ hours	1.3×10^{36}	1.4×10^{34}	1%	2.5×10^{24} kg (about 2.8% of the planet's mass)
Neptune	$7.3 \to 16.1$ hours	1.9×10^{36}	5×10^{31}	0.002%	3.2×10^{24} kg (about 3.1% of the planet's mass)
Jupiter	$9.8 \to 9.9$ hours	The moons carry about 1% of the spin angular momentum	All of it	100%	Not required

Saturn	$10.4 \rightarrow 10.6$ hours	The moons carry about 1.4% of the spin angular momentum	All of it	100%	Not required

• **Uranus and Neptune:** the mass of the regular moons is far too small to carry the angular momentum lost; most of the angular momentum was ejected by water and volatiles that crossed the corotation line. The mass of water that left Uranus (2.5×10^{24} kg) is 270 times the mass of all its regular moons together (9×10^{21} kg). Had this water condensed into moons, the angular momentum would have been kept in the moon system; therefore the water was ejected out into interplanetary space. The full calculation – with the birth radius, compressibility and flattening – is in Paper 11 (Table L.1): 1.6%–10.8% of the mass, water and gas.

• **Jupiter and Saturn:** the corotation line stayed almost static, and the moons alone account for the small transfer of angular momentum, in full agreement with the thick atmospheric cap that blocks the escape of water. Nor do the rings store significant angular momentum: the mass of Saturn's rings is about one part in ten thousand (10^{-4}) of the mass of its moons.

• **Mass return on an active moon:** most of the ice grains ejected in the jets of Enceladus fall back onto its surface as snow, and only a tiny fraction escapes and feeds the E ring [Kempf et al. 2010].

The mechanism of water loss in the distant ice giants

The massive loss of water precisely in the distant ice giants (Uranus and Neptune), and not in the inner gas giants (Jupiter and Saturn), stems from three physical factors:

1. The thickness of the atmospheric cap: distance from the Sun dictated the accretion time of the solid core before the protoplanetary disc dispersed [Paper 11]. Jupiter (90%–97% gas) and Saturn (70%–85% gas) gathered a thick atmospheric cap that confines the water inside, whereas Uranus and Neptune (10%–20% gas) were left with a thin cap that allows free leakage.

2. Escape speed (the depth of the potential well): the escape speed is defined as the minimum energetic threshold required to break free of the planet's gravity. Jupiter ($v_{\rm esc} = 59.5$ km/s) and Saturn (35.5 km/s) are deep potential wells that confine the water, whereas Uranus (21.3 km/s) and Neptune (23.5 km/s) are shallow potential wells that allow water to cross the escape threshold under the same rotational push.

3. Mechanical ejection versus thermal evaporation: the escape of the water was not driven by thermal radiation but by mechanical lifting of mass to the corotation line during the fast-rotation stage (a day of 7–8 hours) and a centrifugal push beyond the line.

Distinguishing the mechanisms of drying: close to the Sun (as on Venus), water is broken up by photolysis from ultraviolet radiation and loses its hydrogen to space; in the distant gas and ice giants, ejection is dictated by the thickness of the atmospheric cap and the depth of the gravitational potential well.

Ratio of the mass of the regular moons to the mass of the parent planet

Planetary system	Mass of the regular moons as a fraction of the parent's mass
Earth	1.2%
Jupiter	0.021%
Saturn	0.025%
Uranus	0.010%
Neptune (without the captured Triton)	0.00005%
Mars	0.000002%

PART E Sorting in the ring

E.1 Sorting by collisions, not by gravitational settling

Inside a planet, material separates by gravitational floating and sinking according to density. In a planetary ring, a dense particle and a light particle at the same orbital radius move at the same Keplerian speed, and therefore do not separate by settling. Structural sorting in the ring is done by momentum transfer in collisions: a massive, rigid body keeps its stable orbit, while light or soft particles receive a kinetic impulse, are thrown aside and are ground.

E.2 Observational findings of collisional sorting

- **"Propeller" structures:** in Saturn's A ring Cassini recorded dozens of propeller-like structures, each with a rigid body about $100\ \mathrm{m}$ across at its centre and, on either side, lanes cleared of material several kilometres long [Tiscareno et al. 2006]. These structures lie in the A ring, beyond the corotation line, where blocks do not fall into the planet and clear a clean path for themselves by pushing light particles aside.

• **Disc thickness:** in Saturn's dense rings, blocks centimetres to metres across are concentrated in a thin vertical layer about $10\ \mathrm{m}$ thick. Tiny particles acquire higher random speeds in collisions and spread into a thicker vertical layer.

• **Size distribution:** the abundance of blocks in the ring falls with their diameter (D) according to a power law:

$$n(D) \propto D^{-3}$$

This distribution is a clear statistical signature of a system of continuous mechanical grinding and crushing [Zebker et al. 1985].

• **The Uranian system:** consists of narrow rings of blocks centimetres to metres in size, separated by bands of fine dust.

E.3 The time element: the ring as a grinding conveyor

The size of a block in the ring serves as a chronological measure of how long it has been in the system: a large block is a young body that has not yet been fully ground. Blocks the size of tiny moons enter the ring, break up in its middle and are ground into dust. According to the mechanism of viscous spreading (D.1), most of the crushed dust flows inward to the parent planet, and a minority is pushed outward. The ring functions as a one-way conveyor in which large blocks enter and dust leaves.

• **A structural prediction:** the highest concentration of large, intact blocks in the ring will be found right next to the region where the ring is fed with material.

PART F The life cycle of a ring

F.1 Migration of the line makes rings disappear

The position of the corotation line depends on the rotation rate of the parent planet. When the planet slows – because of the tidal torque of a massive moon or because of water escaping through the line and carrying angular momentum – the corotation line migrates radially outward. Mass that was previously on the line or above it now finds itself below the new line; the tidal forces brake it and force it to migrate inward and collapse onto the planet. If at the same time the supply of material from the ground stops, the ring system empties and disappears.

F.2 The stages of ring evolution

1. The feeding stage: the corotation line is close to the surface, differentiation is active, and material leaks and is trapped inside the Roche limit. The ring is massive (Saturn today; ancient Mars).

2. The decline stage: internal sorting advances, the crust thickens or the cap leaks, the feeding rate falls, and the corotation line migrates outward. Narrow rings remain below the line (Uranus and Neptune).

3. The disappearance stage: feeding from the ground has stopped and the rest of the ring has collapsed inward. What remains in the system is the moon that crossed the line and the mass that fell onto the parent (Mars: the dust layer and the moon Phobos falling inward [Paper 10]).

- **Preventing a ring on Earth:** no long-lived ring developed on the ancient Earth, because the pear that built the Moon swept the equatorial band in which the ring would have formed and gathered its material into the forming Moon [Paper 10].

F.3 Rings expected to disappear

By the rule for sustaining a ring (B.4), a ring system with no feeding moon and no continuous feeding from the parent's ground is doomed to fade away:

Planet	Rings associated with a moon	Rings without a moon	Status and geodynamic assessment
Saturn	A (Pan, Daphnis), F (Prometheus, Pandora), G (Aegaeon), E (Enceladus)	D, C, B	The D, C and B rings exist only because of active feeding from the ground; the measured ring-rain rate (D.3) shows that without it they would disappear.
Uranus	ε (Cordelia, Ophelia), ν (between Portia and Rosalind), μ (Mab)	$\zeta, 6, 5, 4, \alpha, \beta, \eta, \gamma, \delta$	Expected to disappear. No small moons holding them have been found (an open question in the literature).
Neptune	Adams (Galatea), Le Verrier (Despina)	Galle, Lassell	Expected to disappear.
Jupiter	The whole ring system	—	Persists as long as the inner moons shed dust.
Mars	—	The ancient ring	Disappeared when feeding from the ground stopped; the moons are dry and do not feed.

PART G The ring's mark on the moon

G.1 The morphological identification rule

A completely smooth silicate moon indicates that it stayed for a long time inside a ring system.

Deimos is the smoothest small body observed in the Martian environment. A comparative analysis of the shape models of Ernst et al., carried out with the same method for both Martian moons, shows that the median depth of the depressions on Deimos is 1.9% of their width, against 5.5% on Phobos. 84% of the depressions on Deimos are shallower than 6% of their width, against only 56% on Phobos [Paper 10, J.1]. The orbit of Deimos is almost perfectly circular (eccentricity 0.0003) and equatorial. The combination of a circular, equatorial orbit with smooth topography is a clear signature of a long stay in the heart of the ring. Phobos crossed the lower edge of the declining ring, and was therefore only partly ground and kept its system of grooves. Earth had no long-lived ring, and accordingly the Moon shows a crater-saturated surface.

Among the planets, Deimos is the only case of a completely smooth silicate moon, and its parent planet has no ring today. Smooth icy moons (such as Europa, Enceladus and Triton) are geologically active and renew their surfaces, and are therefore excluded from this rule.

G.2 The effect of the ring's chemical composition

The nature of the morphological change on the moon's surface is dictated by the composition of the ring material:

- **An ice ring (accretion mechanism):** cold ice grains are coated with a layer of frost and stick together on contact at low relative speed [Bridges et al. 1984]. This mechanism built the prominent equatorial ridges of Pan, Atlas and Daphnis, which give them a "saucer" shape [Charnoz et al. 2007].
- **A silicate ring (grinding mechanism):** the ancient Martian ring consisted of condensed vapour and silicate material from the crust. Hard silicate grains act as an abrasive, and therefore Deimos was polished and ground without gaining mass.

G.3 Gravity versus material composition

A massive moon holds, with its gravity field, the material that accumulates on its equator. Among the small moons, self-gravity carries no decisive weight, and the composition of the ring material is what determines whether accretion or grinding takes place:

Moon	Diameter (km)	Density ($\mathrm{kg/m^3}$)	Surface gravity ($\mathrm{m/s^2}$)	Ring environment	Morphological signature
Methone	2.9	310	0.0001	Ice (arc near the G ring)	Smooth, coated
Daphnis	7.6	340	0.0004	Ice (A ring)	Equatorial ridge
Pan	28	400	0.0016	Ice (A ring)	Equatorial ridge
Atlas	30	460	0.0019	Ice (edge of the A ring)	Equatorial ridge
Deimos	12.4	1,471	0.0025	Silicate and vapour	Ground all over
Phobos	22.2	1,860	0.0058	Declining gaseous edge	Partly ground
Iapetus	1,469	1,088	0.22	?	Equatorial ridge (G.6)

Physical assessment: Deimos has stronger self-gravity than Pan and Atlas, and yet mass accumulated on them while Deimos was ground, because of the difference between an accreting ice ring and a grinding silicate ring.

G.4 The ratio of moon diameter to ring thickness

Because the moon and the ring move in the same equatorial plane, the physical interaction is focused on the moon's equator. The ratio between the moon's diameter (D) and the ring's vertical thickness (H) determines the geometry of the signature on the surface:

Geometric ratio	Interaction mechanism	Signature on the moon's surface	Examples in the system
$D < H$	The moon is fully immersed in the ring's thickness	Complete smoothness	Deimos, Methone
$D > H$	The poles stick out above and below the ring	A broad equatorial belt	Pan, Atlas, Daphnis
$D \gg H$	The ring acts as a thin thread around a spherical body	A narrow ridge, or an erased signature	Iapetus (?)

For Deimos, whose initial diameter at exit was $14 - 22\ \mathrm{km}$ [Paper 10, Table 18], complete smoothness requires a ring with a vertical thickness of at least tens of kilometres. A ring made

of condensed vapour and silicate dust (such as the ancient Martian ring) is much thicker than a classical ring of ice blocks, and therefore meets this geometric requirement.

G.5 The speed of crossing the ring

The rate of a moon's tidal migration is proportional to its mass:

- **A small moon:** migrates outward slowly, stays inside the ring for a long time, and acquires a clear morphological signature.
- **A large moon:** is pushed outward quickly, crosses the ring in a short time, and receives a shallow signature that is easily erased.
- **Relative speed at the encounter:** a moon moving nearly in step with the ring particles experiences slow contact that allows gentle accretion, whereas a moon passing the ring particles at a high relative speed experiences enhanced grinding.

G.6 Iapetus: an open question

Along the equator of Iapetus runs a massive topographic ridge about $20\ \mathrm{km}$ wide, up to $20\ \mathrm{km}$ high and about $1,300\ \mathrm{km}$ long. There are two possible geodynamic explanations for its formation:

1. Crossing Saturn's ring: the ridge was carved or accreted when Iapetus crossed Saturn's ring system near its original birth site at a radius of $2.53R$ [Paper 11].

2. Collapse of a sub-ring: the ridge formed from the collapse of a sub-ring that orbited Iapetus itself [Ip 2006; Levison et al. 2011; Dombard et al. 2012].

The second explanation contradicts the rule that moons have no rings (J.1), although it is common in the scientific literature. Deciding between the models is defined as an open question (Part M).

PART H Signatures on the parent planet

H.1 Saturn: direct empirical measurement

During the final dive manoeuvres of the Cassini spacecraft in 2017 (between Saturn's upper atmosphere and the inner edge of the rings), the flow of material falling from the rings to the parent planet was measured directly:

• **The neutral-material channel:** water, methane, ammonia, carbon monoxide, nitrogen and hydrocarbons flow from the ring to the equatorial plane [Waite et al. 2018], together with nanometre-sized silicate grains [Hsu et al. 2018].

• **The ionised-material channel (ionospheric bands):** electrically charged water molecules flow from the ring along Saturn's magnetic field lines and neutralise hydrogen ions (H_3^+) in the upper atmosphere. This process produces bands that are dark in infrared light at latitudes $25^\circ - 55^\circ$ on both sides of the equator [O'Donoghue et al. 2013].

Hence a ring leaves two distinct signatures on the parent planet: neutral material landing at the equator, and charged material landing in symmetric bands at middle latitudes.

H.2 Mars: prediction and search targets

As the ancient Martian ring system dried, the mass stored in it – condensed vapour and silicate crustal components ground to dust – collapsed onto the surface and is the source of the Martian dust [Paper 10]. Four geological search directions follow:

• **Equatorial deposits with no flow source:** the Medusae Fossae Formation is a low-density, thick rock deposit that extends along the equator (in the accepted literature it is explained as a layer of volcanic ash [Ojha & Lewis 2018]).

• **Isotope ratios:** finding hydrogen with a deuterium-to-hydrogen ratio (D/H) that differs from the local geological environment and matches the composition of Phobos and Deimos.

• **Volcanic layers in Mount Sharp:** the ChemCam instrument identified dozens of stones of volcanic composition in Gale crater [Cousin et al. 2017], and a dark, hard marker layer that extends across the structure.

• **Bands at middle latitudes:** if Mars had an active dipole magnetic field while the ring existed, ionised material landed in symmetric bands on both sides of the equator, as in Saturn's mechanism.

H.3 A test that did not pass: the Atacama block

On 25 April 2026 the Curiosity rover drilled into a flat rock block detached from the ground, named Atacama (about $45\ \mathrm{cm}$ across, mass about $13\ \mathrm{kg}$), in Gale crater, about $320\ \mathrm{km}$ from the equator (the drill got stuck in the rock and was released six days later [NASA/JPL 2026]). We examined the possibility that such detached blocks are ring remnants that fell from orbit. The analysis rejected this hypothesis and established that the block belongs to the layered

sulfate unit of Mount Sharp. This test defines the strict criteria for identifying a genuine ring remnant:

Geological measure	Finding in the Atacama block	Criterion required of a genuine ring remnant
Rock type	Layered sediment evaporated from a water solution	Volcanic or spongy material, without water-laid sedimentary layering
Mineral composition	Hydrated sulfate salts	Dry minerals, or minerals that underwent thermal alteration (heating)
Rind morphology	Desiccation cracks	A fusion crust from passage through the atmosphere
Geochemical relation to the surroundings	Composition identical to the layer beneath it	Composition foreign to and different from the local geological layer

PART I The ring and the Sun

Solar radiation affects ring dynamics through three physical mechanisms:

- **The “spokes” phenomenon:** dark bands that appear across Saturn’s B ring, move together with it and vanish within a few hours [Smith et al. 1981]. Their frequency is seasonal and peaks around the equinoxes, when sunlight strikes the ring plane at a shallow angle [Mitchell et al. 2006]; tiny dust particles (micrometre-sized) become electrically charged and levitate electrostatically above the plane of the disc.
- **Acceleration of the ionised rain:** the Sun’s ultraviolet (UV) radiation ionises neutral water molecules in the ring. The resulting ions are trapped in the magnetic field lines and fall to the middle latitudes of the parent planet (H.1); in this way the Sun accelerates the rate at which the ring empties.
- **Radiation drag (the Poynting–Robertson effect):** photons of solar radiation brake the orbital momentum of tiny dust grains and force them to spiral inward into the planet.

PART J Moons and small bodies

J.1 Ruling out rings around moons

No ring system has been found around any moon in the Solar System. The initial report of a suspected ring around Saturn’s moon Rhea was examined observationally and ruled out

completely [Tiscareno et al. 2010]. In the vicinity of a moon the dominant gravity field of the parent planet rules, and therefore any mass ejected from a moon is captured by the planetary dynamics and joins the planet's ring system (B.3).

J.2 Small bodies with rings

Three small bodies in the outer Solar System have dense rings: the centaur Chariklo [Braga-Ribas et al. 2014], and the dwarf planets Haumea [Ortiz et al. 2017] and Quaoar [Morgado et al. 2023; Pereira et al. 2023]. Unlike the rings of the gas and ice giants, all of these rings lie above the corotation line (Table A.3). Four of the five known rings of these bodies lie between $2.01\,a_c$ and $2.18\,a_c$, right next to the radius of the 1:3 spin-orbit resonance:

$$a_{1:3} = 3^{2/3} a_c \approx 2.08\,a_c$$

- **Haumea:** the rotation period is 3.9 hours, and the ends of its long geometric axis ($1,161$ km) extend beyond the position of the corotation line ($1,104$ km) [Paper 11, K.2]. Material released from the ends of the body lies above the line, is pushed radially outward and is locked in the 1:3 resonance. According to the unified mechanism, Haumea's ring is made of material ejected from a body on the verge of an active birth that does not condense into a moon.
- **Quaoar:** the body rotates more slowly, and its surface is far from the corotation line ($3.6R$), so the source of the material of its rings remains open. Nevertheless, its outer ring – lying far beyond the Roche limit and still remaining a ring disc without condensing into a moon – is the strongest observational evidence that a ring does not build moons (C.2).

PART K A systematic review by body

Parent body	State of the ring today	Source of the feeding material	Position relative to the corotation line	Interpretation by the unified mechanism
Earth	None	—	—	The pear swept the equatorial band; no long-lived ring existed.
Mars	None (there was one in the past)	Directly from the ground: vapour and crustal material through the volcano on the protrusion	On the line, which migrated outward	It was fed, pushed Deimos and led Phobos; it disappeared when feeding stopped, 3.7–3.5 billion years ago; the moons are dry [Paper 10].

Jupiter	Tenuous dust	Through moons: Metis, Adrastea, Amalthea, Thebe	Around the line	The thickest atmospheric cap; almost no leakage from the interior. Persists as long as the moons shed dust.
Saturn	Massive, pure water ice	Directly from the ground (D, C, B, A); through moons (F, G, E)	Around the line; the A ring beyond the line, the edge of A at the Roche limit	A steady flow in equilibrium: the leakage rate equals the ring-rain rate. The A ring is the parent's ring that spread beyond the line.
Uranus	Narrow and dark	ε, ν, μ: moons; the rest: without a moon	Below the line $(0.51 - 0.62)$; the μ ring beyond it	A thin cap; the water escaped and the line migrated to $3.26R$. The rings without a moon are expected to disappear.
Neptune	Narrow, with arcs	Adams, Le Verrier: moons; Galle, Lassell: without a moon	Below the line $(0.50 - 0.75)$	The same mechanism as Uranus; the rings without a moon are expected to disappear.
Haumea	A dense ring	The ends of the long axis that cross the line	Above the line $(2.07$)	A body on the verge of birth; the ejected material is locked in the 1:3 resonance.
Chariklo	Two rings	Open	Above the line $(2.10 - 2.18)$	Dynamical stability in the 1:3 resonance.
Quaoar	Two rings	Open	Above the line $(1.25; 2.01)$	Lies outside the Roche limit and does not condense into a moon.

PART L Requirements, predictions and refutation criteria

1. Quantitative and structural requirements of the model

1. In the Saturn system, the rate at which mass leaks from the parent body into the ring equals exactly the rate at which ring rain removes mass (steady state).

2. The isotopic composition of the ring material is identical to that of the parent planet's outer layer, and differs from that of comets or interplanetary dust.

3. No new moon condenses from ring material.

4. Every completely smooth silicate moon indicates a long stay inside a ring system.

2. Scientific predictions for empirical testing

1. The MMX mission (from 2027): on Deimos, ancient, shallow, worn depressions and a chemical composition identical to that of Phobos will be measured; on Phobos, only slight mechanical grinding will be measured.

2. Fast small bodies: small, elongated bodies whose axis ends are close to the corotation line will hold ring systems near the resonance radius $2.08\,a_c$.

3. Size distribution in Saturn's rings: the largest and most intact bodies in the ring will be concentrated in the geometric regions where the ring is fed with material (E.3).

4. Deposits on Mars: equatorial deposits originating in the material of the ancient ring will be found on the surface of Mars, with a chemical and isotopic composition foreign to the local geological layer and matching the composition of Phobos and Deimos.

5. Decline of the moonless rings: in the narrow moonless rings of Uranus and in Neptune's Galle and Lassell rings no feeding moon will be found, and future measurements will record their thinning and disappearance.

3. Strict refutation criteria

1. Direct observation of a moon condensing out of a ring disc.

2. Discovery of a completely smooth silicate moon that never stayed in a ring environment.

3. Measurement of a planetary ring whose isotopic composition contradicts that of the parent planet and matches a source outside the system.

4. Discovery of a massive ring lying entirely far above the corotation line of a large planet, with no orbital resonance stabilising it.

5. Documentation of a ring with no feeding moon and no feeding from the ground that keeps its mass over geological time without thinning.

PART M Open questions for further research

1. How the ridge of Iapetus formed: did the equatorial ridge form by crossing Saturn's ring near the birth site, or by the collapse of a local sub-ring (G.6)?

2. The source of mass at Quaoar and Chariklo: what physical mechanism supplies material to the rings of small bodies whose surfaces are far from the corotation line?

3. How tight the cap is at Jupiter: is there a tiny internal leakage of material at Jupiter in addition to the dust supplied by the inner moons, or does the thick cap seal the body completely?

4. Mapping the ring deposits on Mars: where exactly did the mass of the ancient Martian ring land, and what was its exact vertical thickness?

5. Verifying the feeding rate at Saturn: a direct measurement of the rate at which material leaks from Saturn's interior into the ring, for a quantitative comparison with the measured ring-rain rate.

DATA & CODE Data and code availability

The positions of the corotation line and the Roche limit, the surface gravity of the moons, the emptying and feeding times of Saturn's rings and the data of Figure A.1 were computed with a Python script based on the basic formulas and the data cited in the paper. The code and data are fully reproducible and will be included in the series code package.

REFERENCES References

Braga-Ribas, F., Sicardy, B., Ortiz, J. L., et al. (2014). A ring system detected around the Centaur (10199) Chariklo. Nature, 508, 72–75.

Bridges, F. G., Hatzes, A., & Lin, D. N. C. (1984). Structure, stability and evolution of Saturn's rings. Nature, 309, 333–335.

Burns, J. A., Showalter, M. R., Hamilton, D. P., et al. (1999). The formation of Jupiter's faint rings. Science, 284, 1146–1150.

Charnoz, S., Brahic, A., Thomas, P. C., & Porco, C. C. (2007). The equatorial ridges of Pan and Atlas: terminal accretionary ornaments? Science, 318, 1622–1624.

Callos, S., et al. (2025). A survey of Cassini images of spokes in Saturn's rings. The Planetary Science Journal.

Cousin, A., Sautter, V., Payré, V., et al. (2017). Classification of igneous rocks analyzed by ChemCam at Gale crater, Mars. Icarus, 288, 265–283.

Crida, A., & Charnoz, S. (2012). Formation of regular satellites from ancient massive rings in the Solar System. Science, 338, 1196–1199.

Ćuk, M., Minton, D. A., Pouplin, J. L. L., & Wishard, C. (2020). Evidence for a past Martian ring from the orbital inclination of Deimos. The Astrophysical Journal Letters, 896, L28.

Cuzzi, J. N., & Estrada, P. R. (1998). Compositional evolution of Saturn's rings due to meteoroid bombardment. Icarus, 132, 1–35.

Cuzzi, J. N., Burns, J. A., Charnoz, S., et al. (2010). An evolving view of Saturn's dynamic rings. Science, 327, 1470–1475.

Dombard, A. J., Cheng, A. F., McKinnon, W. B., & Kay, J. P. (2012). Delayed formation of the equatorial ridge on Iapetus from a subsatellite created in a giant impact. Journal of Geophysical Research: Planets, 117, E03002.

Ernst, C. M., Daly, R. T., Gaskell, R. W., et al. (2023). High-resolution shape models of Phobos and Deimos from stereophotoclinometry. Earth, Planets and Space, 75, 103.

Goldreich, P., & Tremaine, S. (1980). Disk–satellite interactions. The Astrophysical Journal, 241, 425–441.

Hsu, H.-W., Schmidt, J., Kempf, S., et al. (2018). In situ collection of dust grains falling from Saturn's rings into its atmosphere. Science, 362, eaat3185.

Iess, L., Militzer, B., Kaspi, Y., et al. (2019). Measurement and implications of Saturn's gravity field and ring mass. Science, 364, eaat2965.

Ip, W.-H. (2006). On a ring origin of the equatorial ridge of Iapetus. Geophysical Research Letters, 33, L16203.

Kempf, S., Altobelli, N., Schmidt, J., et al. (2023). Micrometeoroid infall onto Saturn's rings constrains their age to no more than a few hundred million years. Science Advances, 9, eadf8537.

Levison, H. F., Walsh, K. J., Barr, A. C., & Dones, L. (2011). Ridge formation and de-spinning of Iapetus via an impact-generated satellite. Icarus, 214, 773–778.

Kempf, S., Beckmann, U., & Schmidt, J. (2010). How the Enceladus dust plume feeds Saturn's E ring. Icarus, 206, 446–457.

Lynden-Bell, D., & Pringle, J. E. (1974). The evolution of viscous discs and the origin of the nebular variables. Monthly Notices of the Royal Astronomical Society, 168, 603–637.

Mitchell, C. J., Horányi, M., Havnes, O., & Porco, C. C. (2006). Saturn's spokes: lost and found. Science, 311, 1587–1589.

Morgado, B. E., Sicardy, B., Braga-Ribas, F., et al. (2023). A dense ring of the trans-Neptunian object Quaoar outside its Roche limit. Nature, 614, 239–243.

NASA/JPL (2026). NASA's Curiosity rover frees its drill from a rock (PIA26723); Curiosity takes close look at rock that got stuck on drill (PIA26724).

O'Donoghue, J., Stallard, T. S., Melin, H., et al. (2013). The domination of Saturn's low-latitude ionosphere by ring 'rain'. Nature, 496, 193–195.

O'Donoghue, J., Moore, L., Connerney, J., et al. (2019). Observations of the chemical and thermal response of 'ring rain' on Saturn's ionosphere. Icarus, 322, 251–260.

Ojha, L., & Lewis, K. (2018). The density of the Medusae Fossae Formation: implications for its composition, origin, and importance in Martian history. Journal of Geophysical Research: Planets, 123, 1368–1379.

Ortiz, J. L., Santos-Sanz, P., Sicardy, B., et al. (2017). The size, shape, density and ring of the dwarf planet Haumea from a stellar occultation. Nature, 550, 219–223.

Owen, T., Danielson, G. E., Cook, A. F., et al. (1979). Jupiter's rings. Nature, 281, 442–446.

Pereira, C. L., Sicardy, B., Morgado, B. E., et al. (2023). The two rings of (50000) Quaoar. Astronomy & Astrophysics, 673, L4.

Porco, C. C., Thomas, P. C., Weiss, J. W., & Richardson, D. C. (2007). Saturn's small inner satellites: clues to their origins. Science, 318, 1602–1607.

Sicardy, B., Leiva, R., Renner, S., et al. (2019). Ring dynamics around non-axisymmetric bodies with application to Chariklo and Haumea. Nature Astronomy, 3, 146–153.

Smith, B. A., Soderblom, L., Beebe, R., et al. (1981). Encounter with Saturn: Voyager 1 imaging science results. Science, 212, 163–191.

Thomas, P. C., Burns, J. A., Hedman, M., et al. (2013). The inner small satellites of Saturn: a variety of worlds. Icarus, 226, 999–1019.

Tiscareno, M. S., Burns, J. A., Hedman, M. M., et al. (2006). 100-metre-diameter moonlets in Saturn's A ring from observations of 'propeller' structures. Nature, 440, 648–650.

Tiscareno, M. S., Burns, J. A., Cuzzi, J. N., & Hedman, M. M. (2010). Cassini imaging search rules out rings around Rhea. Geophysical Research Letters, 37, L14205.

Verbiscer, A., French, R., Showalter, M., & Helfenstein, P. (2007). Enceladus: cosmic graffiti artist caught in the act. Science, 315, 815.

Verbiscer, A. J., Skrutskie, M. F., & Hamilton, D. P. (2009). Saturn's largest ring. Nature, 461, 1098–1100.

Waite, J. H., Perryman, R. S., Perry, M. E., et al. (2018). Chemical interactions between Saturn's atmosphere and its rings. Science, 362, eaat2382.

Wisdom, J., Dbouk, R., Militzer, B., et al. (2022). Loss of a satellite could explain Saturn's obliquity and young rings. Science, 377, 1285–1289.

Zebker, H. A., Marouf, E. A., & Tyler, G. L. (1985). Saturn's rings: particle size distributions for thin layer models. Icarus, 64, 531–548.

Duran, Z. (2026). The LEF Model series, Papers 1–14. Zenodo. doi:10.5281/zenodo.22743615 and lef-model.com.

DECLARATIONS Declarations

Use of artificial intelligence. The author used Claude (Anthropic) for calculations, for gathering data from the sources, for building the figure and for drafting. He also used a conversation with Gemini (Google) to examine ideas and data. The ideas and claims of the model are the author's. All data were checked against the cited sources.

Funding and competing interests. This research received no external funding. The author declares no competing interests.

© 2026 Zohar Duran. CC BY 4.0. lef-model.com